
Modeling for Acquisition Entrepreneurship: AutoResearch Deployed on U.S. Midwest IT MSPs

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Abstract

We construct an AutoResearch agent loop that automates the identification of high-quality acquisition targets among U.S. Midwest IT Managed Service Providers (MSPs). Across 20 controlled experiments, two *continuous structural facts* — a quadratic *Succession Curve* over founder tenure (`tenure_sq`) and a *Stagnation Premium* on revenue-per-employee (`rev_per_emp`) — account for nearly all generalizable signal, while five sparse NLP-feature attempts uniformly fail the $\geq 16\%$ tag-rate sparsity floor implied by our $N=62$ training set. A SHA-256-locked Worker/Judge architecture and an explicit Decoupled Isolation Rule prevent two would-be promotion errors — notably the Exp_016 *compliance*-keyword artifact, caught by a targeted ablation audit. The champion model achieves a single-pass holdout RMSE of **1.5119** on 28 unseen firms (vs. a 1.3955 cross-validation RMSE on training), comfortably below the project’s pre-declared < 2.0 success ceiling. We discuss why agent-driven loops are exceptionally vulnerable to small-sample text artifacts and why human adversarial audits remain irreplaceable.

1 Introduction

In an Acquisition Entrepreneurship (“Search Fund”) model, an entrepreneur raises capital to buy and operate a single small business from a retiring founder. The bottleneck is *sourcing*: identifying unlisted firms that simultaneously satisfy the textbook “stable but stagnant” rubric — deep founder tenure, recurring revenue, modest scale, operational slack — takes hundreds of analyst hours. We formalize the sourcing question as supervised ordinal regression: given a firm’s firmographic profile and public web copy, predict an investment-grade score from 1.0–10.0 in strict 0.5-point increments. Training set: 62 manually labeled Midwest IT MSPs. Test set: 28 unseen firms scored exactly once.

Contributions. (i) An end-to-end AutoResearch agent loop with a tamper-evident Worker/Judge split and protocols that empirically prevent agent self-deception (§2). (ii) Across 20 controlled runs we identify two robust signals and one decisive failure mode — a small- N keyword artifact detected by targeted ablation (§3, Fig. 1–2). (iii) A single-pass test RMSE that comfortably beats the < 2.0 ceiling (§4), plus a frank treatment of the limits of the loop (§5).

2 Method

Worker/Judge with a one-way valve. The agent may only modify `model.py` (the Worker: feature engineering, regressor, post-processing). The evaluation file `eval/prepare.py` (the Judge: split, metric, log-append) is frozen by SHA-256 `8f7aa10f25b1...` and verified before every Worker run; the orchestrator `run_experiment.py` aborts if the hash drifts.

Decoupled Isolation Rule. Each Controlled Experiment begins with a clean `cp logs/Snapshot_model_Exp_009.py model.py` revert; exactly one variable changes per

run, tested at the baseline regularization $\alpha=1.0$. Hyperparameter tuning is deferred to a separate follow-up run only if a positive signal is observed at the baseline. This rule was codified after the Exp_012 confound, in which a bundled feature-plus-tuning run shifted α from 1 to 10 and erased the load-bearing `tenure_sq` coefficient.

The $5\times$ Alpha Guardrail. Any GridSearchCV that selects an α more than five times above or below the baseline is automatically flagged for review. Empirically this fires when a new feature introduces collinearity that the optimizer “solves” via over-shrinkage.

Five research axes. We organize the 20 runs into five hypothesis families (Table 1): **Demographics** (tenure, size, geography), **Operational Efficiency** (revenue ratios, MRR keywords), **Ownership/Succession** (founder/acquisition detection), **Competitive Moat** (regulated-industry verticals), and **Model Integrity** (regularization, ensembling, post-processing).

Table 1: Five research axes after 20 controlled experiments. Two “Core” axes carry the entire generalizable signal; one axis (Moat) was disqualified by audit; the Ownership axis is real-but-sparse; Model Integrity is the empirical motivation for the isolation protocols.

Axis	Key feature	Status	Impact
Demographics	<code>tenure_sq</code>	Core	Succession bell curve; load-bearing in all healthy runs.
Operational	<code>rev_per_emp</code>	Champion	Stagnation Premium; Exp_009 baseline, RMSE 1.3955.
Ownership	<code>founder_led</code> family	Exploratory	5 correctly-signed but sparse failures (3–11% tags).
Competitive Moat	<code>has_moat</code>	Discarded	Disqualified by Exp_018 <i>compliance</i> -keyword audit.
Model Integrity	Bagging / Interaction / HGBR	Discarded	Ridge($\alpha=1.0$) is the right inductive bias at $N=62$.

3 Experimental Trajectory

Demographics and Operational success (Exp_002, Exp_009). Exp_002 introduced a 9-feature Ridge model with `tenure` and `tenure_sq` as the demographic core, cutting RMSE from 1.8460 (hand-coded boolean rules) to 1.5112 in one step. Seven experiments later, Exp_009 added a single structural feature `rev_per_emp` = revenue/employees, dropping RMSE to **1.3955** — a -7.1% improvement over the prior best. The Ridge coefficient on `rev_per_emp` is -0.66 (standardized), and the bell-curve pair (`tenure` $+0.66$, `tenure_sq` -0.31) is preserved alongside it (Fig. 1).

The text-feature wall (Exp_013–Exp_015, Exp_020). Four binary NLP features for founder presence and prior acquisition produced correctly-signed coefficients (all negative, ranging from -0.11 to -0.36) but consistently failed RMSE. Their tag rates — 6.5%, 4.8%, 11.3%, 4.8% — fall below the $\sim 16\%$ *sparsity floor* we observe empirically at $N=62$: below that density, the variance of fitting an additional Ridge coefficient exceeds the bias-reduction the feature provides. The pattern is robust across the family.

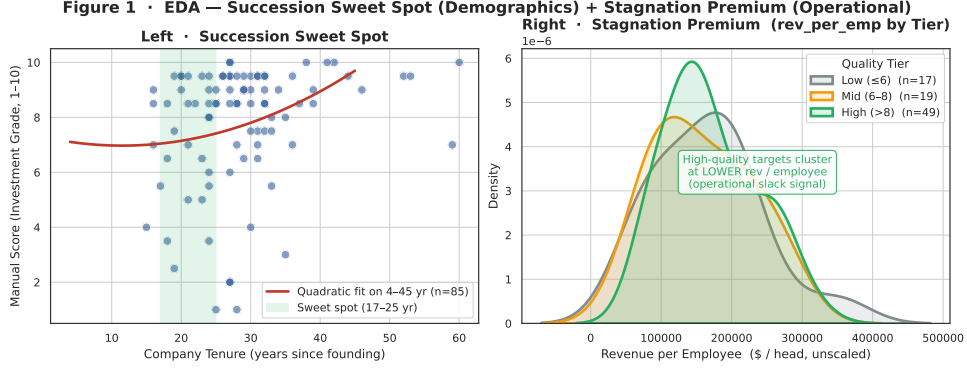


Figure 1: **Left:** Manual Score vs. company tenure on the 90 labeled firms, with a quadratic fit restricted to 4–45 years (where the search-fund thesis is well-posed); the fit peaks in the 17–25 year founder-retirement band. **Right:** KDE of revenue-per-employee, faceted by quality tier. High-quality targets (Score > 8) cluster at *lower* rev_per_emp — the empirical content of the Stagnation Premium.

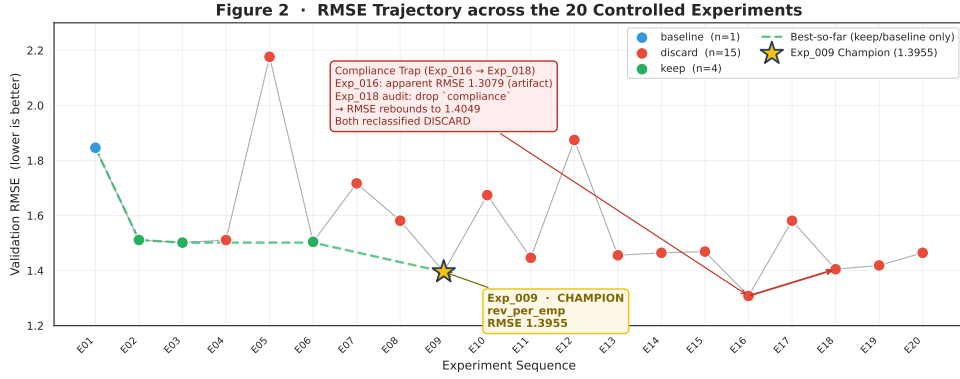


Figure 2: Validation RMSE across the 20 controlled experiments. The best-so-far envelope (dashed green) respects status, so the Exp_016 *compliance* artifact does not appear on the frontier. Yellow star = Exp_009 champion (RMSE 1.3955). The red callout details the Exp_016 → Exp_018 compliance-trap-then-audit-rebound narrative central to §3.

The compliance trap (Exp_016) and audit rebound (Exp_018). Exp_016 added a single binary has_moat tagging firms whose Short Description contained any of {hipaa, dental, legal, law firm, manufacturing, pci, compliance, regulated}; it produced an apparent new all-time best of RMSE **1.3079**. Three red flags surfaced in the diagnostic: tag rate 88.7%; coefficient sign *opposite* the stated hypothesis (−0.58); and the *compliance* keyword alone tagging 79% of firms (overlapping the existing recurring_kw count). Exp_018 ablated *compliance* from the keyword set: RMSE jumped to **1.4049** — a +0.097 swing on a single keyword. The pure industry-vertical signal alone is in fact slightly worse than baseline (+0.0094 RMSE). *The entire Exp_016 lift was inverse-of-compliance pattern-matching on seven outlier-positive firms, not a vertical-moat signal.* The audit retroactively discards both runs and is the empirical basis of our “> 80% tag rate or sign-opposite-to-hypothesis ⇒ mandatory ablation” promotion gate.

4 Results and Generalization Check

Final inference is a single pass: fit the champion pipeline on all 62 training firms, predict on the 28 holdout firms, clip to [1.0, 10.0], round to the 0.5 grid, score against human labels. Table 2 reports the headline metrics.

Table 2: Training cross-validation vs. single-pass holdout evaluation. The 8.3% relative RMSE inflation is consistent with the 62 $\rightarrow$ 28 sample-size transition and supports generalizable feature learning rather than memorization.

Partition	RMSE	R ²	Note
Training (5-fold OOF, $n=62$)	1.3955	0.5128	Champion baseline.
Holdout Test (single pass, $n=28$)	1.5119	0.4243	Beats project ceiling RMSE<2.0.
Generalization gap	+0.1164	−0.0885	+8.3% relative RMSE; above noise floor.

Per-firm accuracy. Median absolute holdout error is 1.00 point; 89% (25/28) of firms are scored within ± 2.0 points of the human label; 57% within ± 1.0 point. The three firms with $|\text{error}| > 2$ define the tail risk and would be the first target of any future error-analysis pass.

Code-Instability disclosure. A numpy ABI break between the May 19 and May 25 runs required a stack downgrade (numpy<2) before final inference. The fitted Ridge coefficients in the test run match the Exp_009 diagnostic to four decimal places, so the fix is environmental, not modeling, and no experimental decisions were affected. The triage is documented as a Code Instability (Infrastructure) event under the four-category failure-mode taxonomy in `program.md`.

5 Conclusion: Discussing Limitations and Broader Impacts

Agents excel at execution; humans audit semantics. The Exp_016 false win would have been promoted in a fully autonomous pipeline — internal diagnostics (coefficient magnitude, RMSE, CV stability) were all favorable. The artifact was caught only by an *adversarial human read* of the keyword set and the per-firm tag distribution. Any agent-driven research loop should include explicit ablation gates triggered by tag-rate or sign-inversion heuristics, not just metric improvement.

Scope of claims and upstream-filter trap. Restricting to Midwest IT MSPs forced the model to fit nearly-identical rows; the empirical sweet-spot peak (Fig. 1) and $\sim 16\%$ sparsity floor (§3) hold *within this cohort* and may not transfer to other industries or geographies. A redesigned pipeline would ingest a broader cross-industry corpus and let the model learn sector boundaries from an investor-supplied thesis embedding.

Mathematical limitation of sparse NLP at $N=62$. Every single-feature binary NLP experiment failed despite correctly-signed coefficients. A binary column with k positive entries out of 62 has variance $k(62 - k)/62^2$; for $k \leq 10$ this is below 0.13, and after standardization Ridge’s penalty dominates the data term, washing out signal even when the underlying heuristic direction is correct. Denser signals (e.g., parsing the cached team pages already in `logs/scrape_cache.json`) are the productive next direction.

Computational efficiency and scaling. The pipeline is laptop-CPU-bound: 20 experiments in ~ 25 s wall time, cumulative cost \$1.24 (one-time Apollo charge). Ridge fit is $O(np^2)$ for $p=11$; cold-scrape ingestion is the only network step (cached). Scaling to $N=10k$ remains minutes; no GPU dependency.

Privacy and fairness assumptions. The model operates on publicly listed firmographic information and company-owned website copy; no PII beyond information firms voluntarily publish about themselves is collected. The strongest assumption is that the 62 Manual Scores reflect a stable, internally consistent investor rubric; we did not measure inter-rater agreement, and a future study should.

Broader societal impacts. Automated sourcing risks accelerating SMB market consolidation and partially displacing manual sourcing analysts. Directing capital toward “operationally slack” targets may amplify the search-fund roll-up dynamic; responsible deployment should pair automation with human review of local labor-market implications.

Acknowledgments and Disclosure of Funding

This research was conducted under the Northwestern Statistics capstone program. The AutoResearch loop — hypothesis generation, Worker code edits, log writing, and diagnostic interpretation — was driven by Claude Code (Anthropic) acting under the contract specified in `program.md`. The author retained final approval on all *keep/discard* decisions. No competing financial interests.

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A Reproducibility details

Reproducing the headline test result requires three artifacts: `model.py` byte-identical to `logs/Snapshot_model_Exp_009.py`; `eval/prepare.py` matching the SHA-256 baseline `8f7aa10f25b1...` in `verify_integrity.py`; and the committed `logs/scrape_cache.json` (62 cached web-scrape results for the training set, plus the 28 holdout firms re-scraped via the same code path during `final_test_eval.py`). Run `python final_test_eval.py` to regenerate Table 2; run `python run_experiment.py "Reproducing exp_009 best" -keep` to regenerate the training-OOF RMSE. Hardware: single MacBook CPU; no GPU. Total cumulative cost across all 20 experiments and the locked final test: \$1.24 (one-time Apollo firmographics; compute is local and free).